

Lynbrook Math Summer Camp 2026

A5 Quadratics and Vieta's Formulas

Lynbrook Math Club

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A5.1 Lecture

Have you come across problems where it asks you to solve for quantities, such as the sum of roots, that seem absurd to actually solve for? Or do you not know how to solve certain problems where the largest exponent of x is 2? This lecture details some approaches.

Quadratics

Definition A5.1.1 (Quadratic). A quadratic function is a sum of powers of x with the highest exponent being 2. For instance, $x^2 + 2x + 1$ is a quadratic.

Definition A5.1.2 (Polynomial). A polynomial of degree n is a sum of powers of x where n is defined as the highest exponent of the powers of x . The degree n must be a natural number $(0, 1, 2, 3, \dots)$.

Using summation notation from **Sequences and Series**, we can write a polynomial of degree n , $P(x)$, as

$$P(x) = \sum_{i=0}^n c_i x^i$$

These c_i are known as *coefficients*.

Definition A5.1.3 (Roots). A root r of a polynomial $P(x)$ is defined as a (real or complex) number that satisfies $P(r) = 0$. In general, for a polynomial of degree n , there can be up to n roots. We can denote each root by r_i .

Now we need a way to find these roots for a quadratic. There are two primary ways to do this, both of which are taught in a standard Algebra 1 course, but we will review them for completeness.

Definition A5.1.4 (Quadratic Factoring). Any standard quadratic $Q(x)$ can be written in standard form as

$$Q(x) = ax^2 + bx + c$$

or in factored form as

$$Q(x) = a(x - r_1)(x - r_2)$$

where a is the same leading coefficient for both.

The most common techniques for factoring quadratics are the following:

1. **Difference of two squares:** $Q(x) = x^2 - a^2 = (x - a)(x + a)$

2. Perfect square trinomial: $Q(x) = x^2 + 2ax + a^2 = (x + a)^2$

The second method to find the roots of a quadratic is the quadratic formula, which is used when it is too hard to find the roots by factoring (use this method as your fallback).

Definition A5.1.5 (Quadratic Formula). The roots r_+ and r_- of a quadratic $Q(x) = ax^2 + bx + c$ can be written as

$$r_{\pm} = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

That is,

$$r_+ = \frac{-b + \sqrt{b^2 - 4ac}}{2a} \quad \text{and} \quad r_- = \frac{-b - \sqrt{b^2 - 4ac}}{2a}$$

are the two roots where we introduced r_+ , r_- , $r_{\pm}$ as notation for roots based on the sign of the square root.

Remark A5.1.6. All quadratics have exactly two roots (if counting multiplicity), and in general a degree n polynomial will have exactly n complex roots.

Vieta's Formulas

Claim A5.1.7 (Factoring Theorem) — Any polynomial $P(x)$ of degree n can be written as

$$P(x) = a(x - r_1)(x - r_2) \cdots (x - r_n)$$

This is known as factoring the polynomial.

Notice that this factored polynomial has the exact same roots as $P(x)$. This is because, plugging in any r_i , the term $(r_i - r_i)$ becomes 0, making the entire product 0. We can adjust the value of a to match the leading coefficient of $P(x)$.

Claim A5.1.8 (Vieta's Formulas for Quadratics) — Vieta's Formulas state that the sum of the roots of a quadratic $ax^2 + bx + c = 0$ is given by

$$r_1 + r_2 = -\frac{b}{a}$$

The product of the roots is given by

$$r_1 r_2 = \frac{c}{a}$$

Let's see why this is true. Suppose we have a quadratic $P(x) = ax^2 + bx + c$. By the factoring theorem, we can write it as $P(x) = a(x - r_1)(x - r_2)$. Expanding this, we get:

$$P(x) = a(x^2 - r_2x - r_1x + r_1r_2) = ax^2 - a(r_1 + r_2)x + a(r_1r_2)$$

Matching the coefficients to a , b , and c , we see that $b = -a(r_1 + r_2)$ and $c = a(r_1r_2)$, which provides Vieta's formulas clearly.

Theorem A5.1.9 (Vieta's Formulas Generalization)

For a polynomial $P(x) = a_n x^n + a_{n-1} x^{n-1} + \cdots + a_1 x + a_0$, the elementary symmetric sums of the roots $r_1, r_2, \dots, r_n$ are given by:

$$\begin{aligned} s_1 &= r_1 + r_2 + \cdots + r_n = -\frac{a_{n-1}}{a_n} \\ s_2 &= r_1 r_2 + r_1 r_3 + \cdots + r_{n-1} r_n = \frac{a_{n-2}}{a_n} \\ &\vdots \\ s_n &= r_1 r_2 r_3 \cdots r_n = (-1)^n \frac{a_0}{a_n} \end{aligned}$$

This means that the sum of the roots for a cubic (degree 3) follows the same pattern as a quadratic. That is, $r_1 + r_2 + r_3 = -\frac{b}{a}$ for $P(x) = ax^3 + bx^2 + cx + d$.

Remark A5.1.10. Rarely do you need to use this entire generalized formula for large n ; usually, you only need to find the sum or product of roots for a quadratic or cubic.

For some extra information on this, you can watch Richard Rusczyk's video here: <https://www.youtube.com/watch?v=OQHnVMWQNfI>